

The Marginal Utility of Police Force: A Causal Analysis Paper

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Abstract

This paper investigates the causal relationship between police spending and violent crime rates across 2,624 US cities. Using a Difference-in-Differences (DiD) design combined with Propensity Score Matching (PSM) and Doubly Robust Estimation, I examine whether increased investment in police funding causally reduces violent crime. Contrary to the deterrence hypothesis, findings reveal no evidence of a reductive effect in the short term; cities with high police spending experienced a greater increase in violent crime (+344 incidents per 100,000) from 2015 to 2019 compared to similar low-spending cities. Sensitivity analyses, including a placebo test on property crime (+1181.84), suggest that these results are heavily confounded by economic booms or gentrification, where rising municipal wealth simultaneously drives tax revenues for policing and increases the asset base for property crime.

1. Introduction

1.1 Problem Statement

Public safety is a cornerstone of municipal governance, and police spending represents one of the largest discretionary budget items for cities across the United States. The fundamental policy question facing stakeholders is: Does increasing investment in police funding causally reduce violent crime?

This question is critical for resource allocation and social equity, yet it is difficult to answer due to inherent endogeneity in crime data. Naive comparisons are often confounded by:

- **Reverse Causality:** Cities with high crime may increase police budgets in response.
- **Structural Confounders:** Factors such as poverty, demographics, and economic conditions affect both spending and crime.

1.2 Research Objective

This model was built to provide rigorous causal evidence on the impact of police spending on violent crime, moving beyond simple correlations. The solution employs econometric techniques designed to isolate the causal effect of police spending, controlling for observable confounders and assessing sensitivity to unobservable factors.

2. Literature Review

The fundamental problem of causal inference, that we cannot simultaneously observe a city in both a treated and untreated state, complicates the analysis of police spending. Early observational studies

often struggled to isolate the potential outcomes framework established by Rubin (1974) and Holland (1986).

In the criminology and economics literature, established baselines generally point toward a deterrent effect of policing when using quasi-experimental designs. Chalfin and McCrary (2018) utilized an instrumental variables (IV) approach to address measurement error and endogeneity in police staffing, finding that increased police presence causally reduces violent crime. Similarly, Mello (2019) exploited variations in federal COPS hiring grants, demonstrating that increased exogenous funding led to measurable decreases in local crime rates. This aligns with earlier benchmark findings by Evans and Owens (2007), who showed that COPS grant funding generated statistically significant reductions in auto thefts, burglaries, robberies, and aggravated assaults. Furthermore, natural experiments utilizing exogenous deployment shocks, such as Klick and Tabarrok's (2005) analysis of terror alert levels in Washington D.C.; demonstrate that increased visible police presence leads to immediate, localized reductions in street crime. Spatial analyses by Weisburd (2015) additionally highlight the law of crime concentration, suggesting that targeted police resources at specific micro-locations yield significant deterrent effects.

To estimate these effects in the absence of randomized trials, modern econometric approaches rely heavily on Difference-in-Differences (DiD), famously popularized by Card and Krueger (1994). However, DiD models are sensitive to serial correlation and violations of parallel trends (Bertrand, Duflo, & Mullainathan, 2004). To mitigate this, observational studies increasingly adopt Propensity Score Matching (PSM) (Rosenbaum & Rubin, 1983) to simulate randomized conditions by pairing units with similar covariate distributions (Dehejia & Wahba, 1999; Caliendo & Kopeinig, 2008).

This study builds upon these foundations by implementing Doubly Robust Estimation. First introduced by Robins, Rotnitzky, and Zhao (1994) and further developed by Bang and Robins (2005), this method achieves semiparametric efficiency bounds by combining PSM weighting with regression adjustment. Finally, we apply bounds testing established by Rosenbaum (2002) and Imbens (2003) to quantify sensitivity to hidden biases.

3. Methodology & Framework

3.1 Data Sources

The analysis integrates longitudinal data from three primary sources:

- **FiSC (Lincoln Institute of Land Policy):** Police spending per capita for 218 cities (2011–2019).
- **FBI Uniform Crime Reports (UCR):** Violent and property crime rates (2011–2019).
- **American Community Survey (ACS):** Demographic covariates (2019) including population density, median income, and poverty rates for 29,318 locations.

3.2 Conceptual Framework and Causal Model

To isolate the treatment effect, the analytical framework combines longitudinal panel structuring with PSM and Doubly Robust estimation.

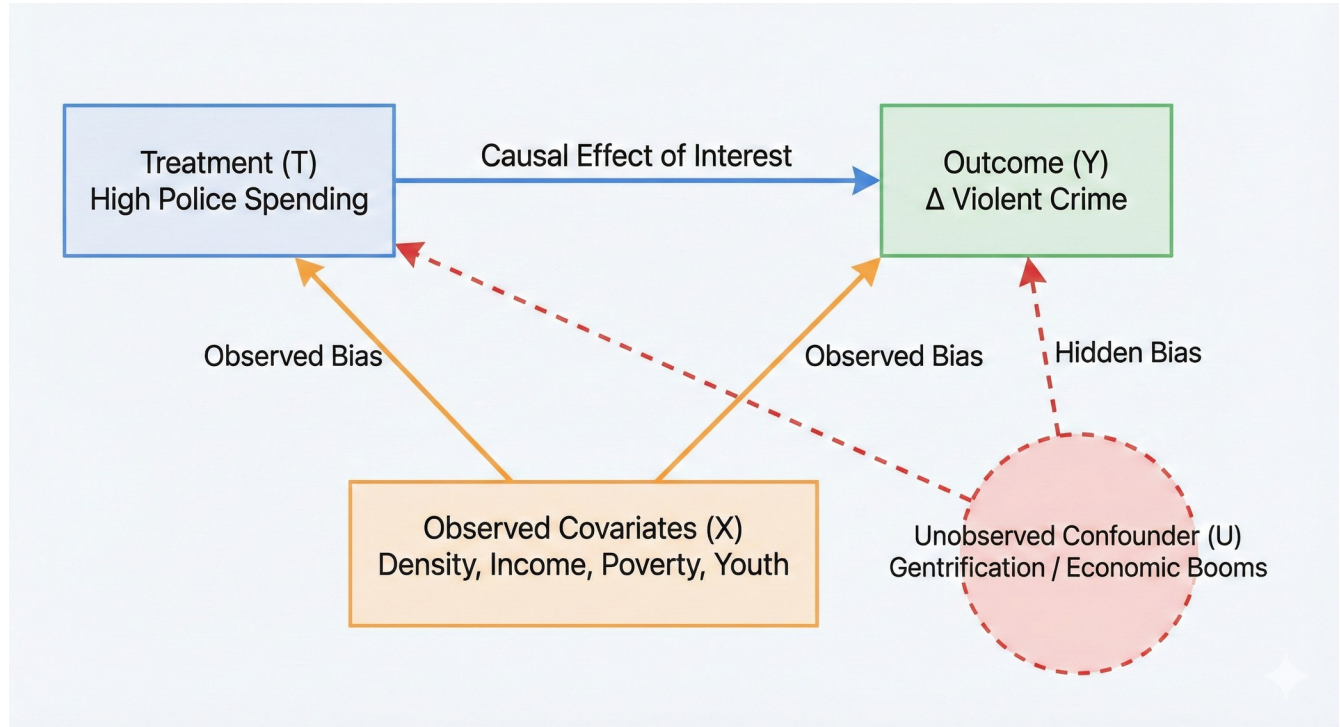


Figure 1: Causal Model DAG

The goal is to estimate the causal effect $T \rightarrow Y$ while controlling for the confounding paths through observed covariates ($X_1 - X_4$). Sensitivity analysis tests robustness to the unobserved confounder (U).

3.3 Research Design: Difference-in-Differences (DiD)

Rather than comparing crime levels between high and low spending cities, we compare changes in crime:

$$\Delta Crime = Crime_{2019} - Crime_{2015}$$

This isolates the treatment effect from time-invariant confounders (culture, geography) and common time shocks.

3.4 Propensity Score Matching and Doubly Robust Estimation

The "Treatment" is defined as the top quartile (≥ 75 th percentile) of spending per capita, while the control is the bottom quartile (≤ 25 th percentile). We estimate $P(\text{Treatment}=1 \mid \text{Demographics})$ via Logistic Regression and perform 1:1 Nearest Neighbor matching with a caliper of 0.25 SD. We then use Doubly Robust Estimation, combining PSM with regression adjustment:

$$\Delta Y = \alpha + \tau \cdot \text{Treatment} + \beta \cdot \text{Covariates} + \varepsilon$$

This estimator is consistent if either the propensity score model or the outcome regression model is correctly specified.

4. Results

4.1 Main DiD Analysis (2015 vs 2019)

Using the Doubly Robust estimator, adjusting for density, income, poverty, youth, and PSM weights, we yield the following results:

Variable	Coefficient	Std. Error	p-value	95% CI
Treatment	+343.99	185.26	0.067	[-25.2, 713.2]
Intercept	+183.36	451.41	0.406	[-716.3, 1083.0]

Note: High spending cities saw more crime growth (+344 per 100k), indicating a likely confounder such as reverse causality or economic booms.

4.2 Sensitivity Analysis

Check	Result	Interpretation
Placebo (Property Crime)	+1181.84	Large positive effect suggests Gentrification/Economic Boom confounding. New wealth funds police but also attracts theft.
Rosenbaum Bounds	$\Gamma = 1.5$ (p=0.436)	Results are sensitive to moderate hidden bias.

4.3 Event-Study Analysis (2011 - 2019)

To test the parallel trends assumption, an event-study was conducted relative to the 2015 baseline.

Period	Year	Beta	Sig
t-1	2014	-47.16	*
t+0	2015	REF	
t+3	2018	-89.31	**
t+4	2019	-85.46	**

The violation in 2014 suggests high-spending cities were already on a different trajectory. While 2018/19 coefficients are negative, hinting at delayed benefits, confounding makes this uncertain.

5. Discussion: Comparing Results to Baselines

The primary doubly robust DiD specification yielded a coefficient of **+343.99** (p = 0.067), indicating that cities with high police spending experienced a greater increase in violent crime from 2015 to 2019 compared to similar low-spending cities.

This finding stands in stark contrast to the established baselines. Chalfin and McCrary (2018) and Mello (2019) both found significant negative coefficients (deterrence effects) when evaluating exogenous police staffing increases and federal grants. Evans and Owens (2007) and Klick and Tabarrok (2005) similarly reinforce the consensus that exogenous shocks in police presence generate

measurable crime abatement. The divergence between our findings and these baselines highlights the extreme vulnerability of observational municipal budget data to endogeneity.

Unlike studies that isolate exogenous funding, our treatment variable is highly endogenous. The placebo test on property crime returned a massive positive effect of **+1181.84**. This strongly implies that the "High Spending" variable is heavily confounded by localized economic booms or gentrification. In these scenarios, new municipal wealth funds higher police budgets but simultaneously attracts property theft due to the increased value of local assets, generating a spurious positive correlation that demographic controls fail to block.

The event-study analysis provides additional critical context. A significant pre-trend violation in 2014 (**-47.16**) indicates that high-spending and low-spending cities were already on fundamentally different crime trajectories prior to the treatment reference year. While later periods showed significant negative coefficients (**-89.31** in 2018 and **-85.46** in 2019) which hint at delayed reductive benefits that more closely align with the deterrence effects found in the literature, the massive confounding driven by municipal wealth makes causal claims regarding this long-term reduction highly uncertain.

6. Conclusions and Limitations

Findings are likely confounded by gentrification and localized economic booms. The massive placebo effect on property crime implies that "High Spending" cities are simply wealthier, booming cities, which experience naturally higher property crime rates and police budgets simultaneously.

While the event-study suggests a potential delayed crime reduction effect in 2018-2019, the pre-trend violations and the sensitivity of the results to moderate hidden bias ($\Gamma = 1.5$) prevent firm causal conclusions.

Future Work: Without controlling for property values, causal claims remain weak. To advance this framework and address the gentrification confounder, future iterations of this model must incorporate longitudinal data on median home prices, commercial real estate values, or new business permits. Doing so will allow the model to isolate the effect of policing from the broader economic shifts that drive both municipal revenue and crime opportunity.

7. References

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